

Honors Class05 Activity: Practice Quiz Questions

Team Members: _____

Learning Objectives

- Test your understanding
- Prepare for Quiz 1

This activity is practice for the first quiz. Answer the questions based on your programming experience. If you are stuck, talk with a team member. If you are *really* stuck, then try the code in a Jupyter notebook, but keep in mind that you won't be able to do this on the quiz.

Problem 1: What is the output?

```
a = "I'm not arguing."  
b = "I'm just explaining why I'm right."  
print(a + b)
```

Problem 2: What is the output?

```
winter_months = ["December", "January", "February", "March"]  
print(winter_months[3])
```

Problem 3: What is the output?

```
x = 2**3 / 8 + 3  
print(x)
```

Problem 4: What is the output?

```
# What do you call a fish with no eyes?  
answer = "fish".replace("i", "")  
answer
```

Problem 5: Explain the error.

```
my_tuple = ("Dr. Nyquist", "Geophysicist", "CST")  
my_tuple[2] = "Earth & Environmental Science"
```

TypeError: 'tuple' object does not support item assignment

Problem 6: What is the output?

```
import numpy as np  
a = 3  
b = 4  
c = np.sqrt(a**2 + b**2)
```

Problem 7: What is the output?

```
np.arange(1, 7)
```

Problem 8: What is the output?

```
len("Data Science")
```

Problem 9: What is the output?

```
from datascience import *  
  
x = make_array(1, -3, 12, -7, 0, -50)  
max(abs(x))
```

Problem 10: What is the output?

```
colors = ["red", "blue", "green"]  
colors.append("yellow")  
print(colors)
```

The next 5 questions are based on the following data table named `dice_results`.

Team	Rolls	Average Sum	Number of 7s	Avg - 7
Team A	50	6.4	7	-0.6
Team B	50	7.1	9	0.1
Team C	50	6.8	8	-0.2
Team D	50	7.3	11	0.3
Team E	50	6.6	6	-0.4
Team F	50	7.0	8	0.0
Team G	50	7.5	12	0.5
Team H	50	6.9	9	-0.1

Simulated dice roll results for eight teams, each rolling 50 times

Problem 11: What is the output?

```
import numpy as np

np.max(dice_results.column("Number of 7s"))
```

Problem 12: What is the output?

```
dice_results.take(2)
```

Problem 13: What is the output?

```
dice_results.where("Average Sum", are.below(6.5))
```

Problem 14:

Write Python code to find the average number of 7s.

Problem 15:

Write Python code to sort the table by Avg - 7.

Design Your Own Quiz Questions

Work with your team to design up to five quiz questions. If you design good ones, you may see them on the actual quiz!